

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Machine Learning

ASSESSMENT TOOL 2 – Scenario-Based Assignment (5 QUESTIONS)

Course Code:	MLA0407	Course Title:	Deep Learning
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Assignment
Weightage:	35% of CIA	Total Marks:	(5 Questions × 10 Marks)
CO1 Statement:	Apply the fundamental concepts of machine learning and neural networks to design and analyze learning algorithms using perceptrons, multilayer perceptrons, forward propagation, backpropagation, gradient descent optimization techniques, and Maximum Likelihood Estimation for solving real-world problems.		
Student Name:	CH. Durga Prasad	Reg. No.:	192425305
Section / Batch:	2024	Date:	06.08.2026

Code of Conduct

I CH. Durga Prasad (192425305) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: _____

QUESTIONS: 5

Scenario-Based Assignment

Total Marks: 50

Annexure A

Duration: 1hr

Deep Learning – Assessment 2

1. Machine Learning vs Deep Learning

A healthcare company wants to develop an AI system to detect pneumonia from chest X-ray images. Initially, the development team plans to use traditional Machine Learning with manually extracted image features. However, after collecting millions of medical images, they decide to switch to a Deep Learning approach.

Question:

Compare Machine Learning and Deep Learning in the context of this scenario. Justify why Deep Learning is a better choice for this application by discussing data requirements, feature extraction, computational resources, accuracy, and real-world implementation challenges.

2. Representation Learning

A retail company has millions of customer purchase records and wants to automatically identify customer buying patterns without manually selecting important features. The company plans to use a Deep Learning model that can learn meaningful representations directly from raw transactional data.

Question:

Explain the concept of Representation Learning and discuss how it helps solve the company's problem. Compare it with traditional feature engineering and explain its advantages in handling complex real-world datasets.

3. Width and Depth of Neural Networks

A startup is developing a handwritten digit recognition system. Two engineers propose different neural network architectures. One suggests using a very wide network with a single hidden layer containing thousands of neurons, while the other recommends a deeper network with multiple hidden layers and fewer neurons per layer.

Question:

Analyze both approaches by explaining the concepts of width and depth in neural networks. Discuss how each architecture affects feature learning, computational complexity, model performance, and generalization. Recommend the most suitable architecture with proper justification.

4. Activation Functions

A company is building a Deep Learning model for image classification. During training, the team observes problems such as slow convergence, vanishing gradients, dead neurons, and unstable outputs in

the final classification layer.

Question:

Evaluate the suitability of Sigmoid, Tanh, ReLU, Leaky ReLU, ELU, and Softmax activation functions for different layers of the neural network. Explain which activation function should be used in each stage of the model and justify your choices based on their characteristics.

5.Unsupervised Training of Neural Networks

A social media platform collects billions of unlabeled user images every month. Since manually labeling such a massive dataset is expensive and time-consuming, the company wants to train a neural network that can automatically discover hidden patterns and useful representations.

Question:

Explain how unsupervised training of neural networks can address this problem. Discuss the learning process, advantages over supervised learning in this scenario, and describe suitable techniques that can be applied for representation learning.

1. Machine Learning vs Deep Learning

Scenario:

A healthcare company wants to detect pneumonia from chest X-ray images. Initially, it uses Machine Learning with manually extracted features. After collecting millions of images, it switches to Deep Learning.

Answer:

Machine Learning (ML) and Deep Learning (DL) are two approaches used to build intelligent systems. ML relies on manually selected features, whereas DL automatically learns features from raw data using neural networks.

Aspect	Machine Learning	Deep Learning
Data Requirement	Works with small or medium datasets	Requires large datasets
Feature Extraction	Manual feature engineering	Automatic feature learning
Computational Resources	Uses CPU and less memory	Requires GPU/TPU and high memory
Accuracy	Moderate accuracy	Higher accuracy for image tasks
Training Time	Faster	Longer training time

In this scenario, Deep Learning is the better choice because the company has **millions of chest X-ray images**. A Convolutional Neural Network (CNN) can automatically learn image features such as edges, lung structures, and pneumonia patterns without manual intervention. Although DL requires more computational power and training time, it provides higher accuracy and better performance in medical image analysis.

Conclusion:

Deep Learning is more suitable because it automatically extracts features, handles large datasets efficiently, and achieves better diagnostic accuracy than traditional Machine Learning.

2. Representation Learning

Scenario:

A retail company wants to identify customer buying patterns from millions of purchase records without manually selecting features.

Answer:

Representation Learning is a Deep Learning technique in which the model automatically learns useful features from raw data instead of relying on manually engineered features.

For example, instead of manually selecting features like customer age, income, or number of purchases, the model automatically learns customer preferences, shopping habits, and relationships between products.

		Advantages
Traditional Feature Engineering	Representation Learning	
Features are manually selected	Features are learned automatically	
Requires domain expertise	Requires little manual effort	
Time-consuming	More efficient for large datasets	
Limited feature discovery	Learns complex hidden patterns	

- Automatically discovers customer buying behavior.
- Handles millions of transaction records efficiently.
- Improves recommendation systems.
- Reduces manual feature engineering.
- Learns complex relationships among products and customers.

Conclusion:

Representation Learning helps the retail company analyze large customer datasets more effectively by automatically learning meaningful patterns and improving business decisions.

3. Width and Depth of Neural Networks

Scenario:

A startup is developing a handwritten digit recognition system. One engineer suggests a wide network, while another suggests a deep network.

Answer:

The **width** of a neural network refers to the number of neurons in each hidden layer, while the **depth** refers to the number of hidden layers.

Wide Network

- One hidden layer with many neurons.
- Easier and faster to train.
- Learns simple features.
- May require many parameters.
- Less effective for complex image recognition.

Deep Network

- Multiple hidden layers with fewer neurons.
- Learns hierarchical features such as edges, curves, and complete digit shapes.
- Higher accuracy for image classification.
- Better generalization on unseen data.

Aspect	Wide Network	Deep Network
Feature Learning	Simple	Hierarchical
Accuracy	Moderate	High
Memory Usage	High	Moderate
Generalization	Lower	Better

Recommendation:

A **deep neural network** is recommended because handwritten digit recognition requires learning multiple levels of image features, resulting in better accuracy and improved generalization.

4. Activation Functions

Scenario:

During image classification model training, problems such as vanishing gradients, dead neurons, and unstable outputs occur.

Answer:

Activation functions introduce non-linearity into neural networks and help the model learn complex patterns.

Sigmoid

- Output range: 0 to 1
- Suitable for binary classification output.
- Suffers from vanishing gradient.

Tanh

- Output range: -1 to 1
- Zero-centered.
- Still affected by vanishing gradients.

ReLU

- Outputs positive values and zero for negatives.
- Fast training.
- Reduces vanishing gradient.
- May cause dead neurons.

Leaky ReLU

- Allows a small negative output.
- Prevents dead neuron problem.
- Better than ReLU in some cases.

ELU

- Produces smooth negative outputs.
- Faster convergence.
- Improves training stability.

Softmax

- Converts outputs into probabilities.
- Used in the final layer for multi-class classification.

Activation Function	Best Use
ReLU	Hidden layers
Leaky ReLU	Hidden layers (avoids dead neurons)
ELU	Deep hidden layers
Sigmoid	Binary output layer
Softmax	Multi-class output layer

Conclusion:

Use **ReLU** for most hidden layers, **Leaky ReLU** or **ELU** if dead neurons occur, and **Softmax** in the final layer for image classification.

5. Unsupervised Training of Neural Networks

Scenario:

A social media platform collects billions of unlabeled images and wants to learn useful patterns without manual labeling.

Answer:

Unsupervised learning trains a neural network using **unlabeled data**. Instead of predicting labels, the model discovers hidden patterns and meaningful representations.

The learning process involves feeding raw images into the network, which automatically learns common features such as edges, textures, shapes, and object similarities.

Advantages

- No need for expensive manual labeling.
- Can utilize billions of unlabeled images.
- Learns useful feature representations.
- Improves image search and recommendation systems.
- Reduces annotation cost and time.

Suitable Techniques

- **Autoencoders:** Learn compressed image representations.
- **Variational Autoencoders (VAEs):** Learn meaningful latent features.
- **Self-Supervised Learning:** Creates labels from the data itself.
- **Contrastive Learning:** Learns similarities between images.
- **Clustering:** Groups similar images together.

Conclusion:

Unsupervised learning is ideal for this scenario because it can learn useful representations from massive unlabeled datasets, reducing labeling costs while improving downstream tasks such as image classification and recommendation.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

